Statements, Logic, and Sets

Mathematics for Engineers I · Module 62765 · Chapter 1

From Fundamentals to AI Apocalypse · personal study guide

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Mathematical arguments use a small grammar of statements, logical connectives, and sets. This chapter builds that grammar: first decide when statements are true, then describe and combine collections of objects precisely.

Start from memory

Try these before reading. If you are unsure, write a guess; the gap tells you what to look for.

1. When is an implication $A \Rightarrow B$ false?
2. Does mathematical "or" allow both statements to be true?
3. How do you negate "for every x, P(x)"?
4. What does $A \subset B$ say about the elements of A?
5. How is $A \times B$ different from the set containing A and B?

Answers: see "Start from memory" under Answers at the end.

How to use this guide

Read one section, hide the page, then reconstruct the rules and examples from memory. Use the answers only after you have committed to a response.

1. **Day 0:** Do the cold start, read §§1.1–1.2, and try the worked examples with the answers covered.
2. **Day 1:** Rebuild the truth table and the set operations from memory; solve practice 1–8.
3. **Day 3:** Solve practice 9–16 without notes, then check each answer and revisit its section reference.
4. **About day 7:** Solve practice 17–24 in mixed order and explain the De Morgan rules aloud.
5. **About day 14:** Reconstruct the derivations, then take one exam problem under time limits.
6. **Before the exam:** Complete the self-check and both exam problems. Return to any skill you cannot demonstrate unaided.

When stuck: identify the definitions involved, test a small example, and write each membership or truth condition explicitly before simplifying.

How the chapter fits together

The chapter moves from evaluating statements to describing collections and proving relationships between them.

Measured · the starting language

- **Statements** §1.1

  true = 1 · false = 0

  Rests on: deciding whether a sentence has a definite truth value.

- **Connectives** §1.1

  $\neg$, $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$

  Rests on: truth values for the component statements and the connective's rule.

Defined · objects and relations

- **Quantifiers** §1.1

  $\forall x\ P(x)$ · $\exists x\ P(x)$

  Rests on: a domain of possible objects and a predicate whose truth can be checked.

- **Sets and membership** §1.2

  $x \in A$

  Rests on: a well-defined collection and a membership test.

- **Set operations** §1.2

  $\cup$, $\cap$, $\setminus$, $\times$, A^c

  Rests on: membership in the original sets and, for a complement, a fixed universal set.

Derived · rules from definitions

- **Logical equivalences** §1.1

  truth tables prove equivalence

  Rests on: matching truth values in every row.

- **Set algebra** §1.2

  compare membership conditions

  Rests on: showing that both sides contain exactly the same elements.

Question	Tool
When is a compound statement true?	Truth table
Does a set equation hold?	Element-by-element membership argument
What is the negation of a quantified claim?	Swap $\forall$ and $\exists$, then negate the predicate

1.1 Propositional Logic

Logic gives precise rules for combining claims. It lets you test arguments, simplify conditions, and express properties that will reappear throughout mathematics.

A statement has a truth value

A **statement** is a sentence that is either true or false; its truth value is 1 or 0, respectively. "$9 > 8$" is a statement. "Friday the 13th is unlucky" is not a statement here because it does not have a definite mathematical truth value. A claim may still be a statement even if its truth is not currently known.

Logical connectives

$$\neg A$$
$$A \wedge B$$
$$A \vee B$$
$$A \Rightarrow B$$
$$A \Leftrightarrow B$$

Truth tables define the meaning of each connective. The notes use 1 for true and 0 for false; $\vee$ is inclusive "or".

Form	Meaning	True when
$\neg A$	not A	A is false
$A \wedge B$	A and B	both A and B are true
$A \vee B$	A or B	at least one is true; both may be true
$A \Rightarrow B$	A implies B	A is false or B is true
$A \Leftrightarrow B$	A iff B	A and B have the same truth value

The implication $A \Rightarrow B$ is false in exactly one case: A is true and B is false. This is why a false premise makes the implication true under the truth-table definition; it does not claim that the conclusion follows from a false premise in ordinary causal language.

Worked example · build a truth table

Test whether $A \Rightarrow B$ has the same truth values as $\neg A \vee B$.

1. List all input pairs: (1,1), (1,0), (0,1), (0,0).
2. Evaluate $\neg A \vee B$: 1, 0, 1, 1.
3. Evaluate $A \Rightarrow B$: 1, 0, 1, 1.
4. The columns match in every row, so the two forms are equivalent. §1.1

Tautologies and logical rules

A **tautology** is true for every assignment of truth values. Two forms are logically equivalent when their truth values agree in every row. A truth table is a direct proof for expressions with a small number of variables.

Rules to recognise and prove with truth tables

$$\neg\neg A \Leftrightarrow A$$
$$A \wedge B \Leftrightarrow B \wedge A$$
$$A \vee B \Leftrightarrow B \vee A$$
$$A \wedge (B \wedge C) \Leftrightarrow (A \wedge B) \wedge C$$
$$A \vee (B \vee C) \Leftrightarrow (A \vee B) \vee C$$
$$A \wedge (B \vee C) \Leftrightarrow (A \wedge B) \vee (A \wedge C)$$
$$A \vee (B \wedge C) \Leftrightarrow (A \vee B) \wedge (A \vee C)$$
$$\neg(A \wedge B) \Leftrightarrow \neg A \vee \neg B$$
$$\neg(A \vee B) \Leftrightarrow \neg A \wedge \neg B$$

These include double negation, commutativity, associativity, distributivity, and De Morgan's laws (Theorem 1.4, notes p. 4).

Trap: the converse $B \Rightarrow A$ is not generally equivalent to $A \Rightarrow B$. "If it rains, the street is wet" does not imply "if the street is wet, it rained."

Predicates and quantifiers

A **propositional form** $P(x)$ depends on variables; assigning values turns it into a statement. A quantifier specifies how many values in a domain must satisfy it.

Quantifiers

$$\forall x\, P(x) : P(x) \text{ holds for every } x \text{ in the domain}$$
$$\exists x\, P(x) : P(x) \text{ holds for at least one } x$$
$$\exists! x\, P(x) : P(x) \text{ holds for exactly one } x$$

The domain matters. Quantifier order matters too: ∃D ∀P R(D,P) is not the same as ∀P ∃D R(D,P).

Negation changes "every" to "there exists at least one counterexample", and changes "there exists" to "none exist".

Negating quantified statements

$$\neg(\forall x\, P(x)) \Leftrightarrow \exists x\, \neg P(x)$$
$$\neg(\exists x\, P(x)) \Leftrightarrow \forall x\, \neg P(x)$$

Negate the predicate after swapping the quantifier.

Worked example · negate a quantified claim

Negate: "For every real number x, $x^2 \geq 0$."

1. Swap ∀ for ∃.
2. Negate the predicate $x^2 \geq 0$ to $x^2 < 0$.
3. The negation is "There exists a real number x with $x^2 < 0$." It is false, so the original claim is true. §1.1

Memory hook: to disprove a universal claim, give one counterexample. To prove an existential claim, give one witness.

Why the grammar matters

The notes connect propositional logic to digital circuits and satisfiability problems. A circuit implements a truth table; a SAT solver asks whether some assignment makes a propositional formula true. These are applications of the truth rules, not extra operators.

Recall check · answer without looking

1. Which row makes $A \Rightarrow B$ false?
2. How can you prove two forms equivalent with a truth table?
3. What is the negation of $\forall x\, P(x)$?

Answers: see "1.1 Propositional Logic" under Answers at the end.

1.2 Set Theory

Sets let us collect objects by a precise membership rule. Logical connectives reappear here as operations on membership conditions.

Membership, descriptions, and inclusion

A **set** is a well-defined collection of distinct objects. Write $x \in A$ when x is a member and $x \notin A$ otherwise. A set may be given by listing its elements, such as $\{1,2,3\}$, or by a property, such as $\{x \in \mathbb{R} : x > 0\}$. Order and repetition do not change a set.

Subset and equality

$$A \subset B \Leftrightarrow \text{every } x \in A \text{ also belongs to } B$$
$$A = B \Leftrightarrow A \subset B \text{ and } B \subset A$$
$$A \subsetneq B \Leftrightarrow A \subset B \text{ and some element of } B \text{ is not in } A$$

The notes use $\subset$ for subset and $\subsetneq$ for proper subset. Some authors use $\subseteq$ for subset.

The empty set $\varnothing$ has no elements and is a subset of every set. Standard number sets are $\mathbb{N}$ (positive natural numbers in these notes), $\mathbb{Z}$ (integers), $\mathbb{Q}$ (rationals), $\mathbb{R}$ (reals), and $\mathbb{C}$ (complex numbers).

Intervals, power sets, and size

Intervals describe parts of the real line. Round endpoints are excluded and square endpoints included: (a,b), $[a,b)$, $(a,b]$, $[a,b]$. Unbounded intervals include (a,∞) and $(-\infty,b]$; the notes also use $\mathbb{R}=(-\infty,\infty)$, $\mathbb{R}^+=(0,\infty)$, and $\mathbb{R}_{\geq 0}=[0,\infty)$. The power set $\mathcal{P}(X)$ contains every subset of X. For finite X, cardinality $|X|$ counts its elements, and $|\mathcal{P}(X)| = 2^{|X|}$.

Worked example · enumerate a power set

Let $X=\{a,b\}$. List $\mathcal{P}(X)$ and its size.

1. Include the empty subset: $\varnothing$.
2. Include the two one-element subsets: $\{a\}$, $\{b\}$.
3. Include the whole set: $\{a,b\}$. Therefore $\mathcal{P}(X)=\{\varnothing,\{a\},\{b\},\{a,b\}\}$ and $|\mathcal{P}(X)|=4=2^2$. §1.2

Set operations

Each operation is easiest to understand by asking when an arbitrary element x belongs to the result. Sets with empty intersection are called **disjoint**.

Operations and membership tests

$$x \in A \cup B \Leftrightarrow (x \in A) \vee (x \in B)$$
$$x \in A \cap B \Leftrightarrow (x \in A) \wedge (x \in B)$$
$$x \in A$$
$$, B \Leftrightarrow (x \in A) \wedge (x \notin B)$$
$$A \times B = \{(a, b) : a \in A \,\text{and}\, b \in B\}$$
$$A^c = X$$
$$, A, \text{ where } X \text{ is the fixed non} - \text{empty universal set}$$

Pairs are ordered: (a,b) is generally different from (b,a). A complement is defined relative to X.

Union collects elements in either set; intersection keeps elements in both; difference keeps elements in A but removes those in B. The Cartesian product contains ordered pairs, not sets of the factors.

Operations on n sets

$$\bigcup_{k=1}^{n} A_k \text{ contains x when x belongs to at least one } A_k$$
$$\bigcap_{k=1}^{n} A_k \text{ contains x when x belongs to every } A_k$$
$$\prod_{k=1}^{n} A_k = A_1 \times \cdots \times A_n, \text{ the ordered n} - \text{tuples } (x_1, ..., x_n) \text{ with } x_k \in A_k$$

In particular, $\mathbb{R}^2$=$\mathbb{R}\times\mathbb{R}$ is the plane and $\mathbb{R}^3$=$\mathbb{R}\times\mathbb{R}\times\mathbb{R}$ is three-dimensional space.

Worked example · combine operations

Let A={1,2,3} and B={3,4,5}. Find A∪B, A∩B, and A\B.

1. Membership in at least one gives A∪B={1,2,3,4,5}.
2. Membership in both gives A∩B={3}, so the sets are not disjoint.
3. Members of A that are not in B give A\B={1,2}. §1.2

Set algebra and De Morgan's laws

Set identities can be proved by translating both sides into membership conditions. For example, $x \in (A \cap B)^c$ exactly when $x \notin A \cap B$; this means it is not true that x is in both sets, or equivalently $x \notin A$ or $x \notin B$. Thus $x \in A^c \cup B^c$, proving $(A \cap B)^c = A^c \cup B^c$.

Set-algebra rules (Theorem 1.13)

$$(A^c)^c = A$$
$$A \cap B = B \cap A$$
$$A \cup B = B \cup A$$
$$A \cap (B \cap C) = (A \cap B) \cap C$$
$$A \cup (B \cup C) = (A \cup B) \cup C$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$(A \cap B)^c = A^c \cup B^c$$
$$(A \cup B)^c = A^c \cap B^c$$

Complements use the same fixed universal set X.

Trap: $A \setminus B$ is not generally $B \setminus A$. Check the direction by testing membership: the first set supplies candidates; the second removes them.

Recall check · answer without looking

1. How do you prove A=B from the definition?
2. What is the difference between $A \cap B$ and $A \times B$?
3. What information is needed before A^c is defined?

Answers: see "1.2 Set Theory" under Answers at the end.

Formula map and notation

Idea	Rule	Check
Implication	$A \Rightarrow B \Leftrightarrow \neg A \vee B$	False only for A=true, B=false
Negated quantifier	$\neg \forall x P(x) \Leftrightarrow \exists x \neg P(x)$	One counterexample is enough
Union/intersection	$\cup$ means or; $\cap$ means and	Translate to membership
Difference	$A \setminus B$: in A and not in B	Order matters
Complement	$A^c = X \setminus A$	State X
Power set	$\lvert\mathcal{P}(X)\rvert = 2^{\lvert X\rvert}$, finite X	Includes $\varnothing$ and X

Check every answer: for logic, test all truth assignments; for sets, test membership element by element and state the universal set when using complements.

Derive it cold

Rebuild these on blank paper before opening the answers in the relevant sections.

1. Use the truth table to derive $A \Rightarrow B \Leftrightarrow \neg A \vee B$. §1.1
2. Use the truth table to derive both logical De Morgan laws. §1.1
3. Negate a universal and an existential statement by swapping quantifiers and negating the predicate. §1.1
4. Prove one set identity by writing an arbitrary x-membership condition for each side. §1.2
5. Derive the set De Morgan laws from the logical De Morgan laws and membership tests. §1.2

Explain it

Answer each "why" in your own words before checking the model response.

1. Why is $A \Rightarrow B$ true when A is false under the truth-table definition?
2. Why does matching truth-table columns prove equivalence?
3. Why can one counterexample disprove a universal statement?
4. Why does quantifier order change the meaning of a claim about dogs and people?
5. Why must the universal set be fixed to define a complement?
6. Why is $A \setminus B$ different from $B \setminus A$?
7. Why are ordered pairs in $A \times B$ generally not the same as those in $B \times A$?
8. Why can a set identity be proved by comparing membership conditions?

Answers: see "Explain it" under Answers at the end.

Mixed practice

Try all 24 problems before opening the answer key. The sections alternate so you must identify the right tool.

1. For A=1 and B=0, evaluate $\neg$A, A$\wedge$B, A$\vee$B, A$\Rightarrow$B, and A$\Leftrightarrow$B.
2. Let A={1,2,3}, B={3,4}. Find A$\cup$B, A$\cap$B, A\B, and B\A.
3. Is "There are infinitely many twin primes" a statement even if its truth is unproven? Explain briefly.
4. Let X={a,b}. List $\mathcal{P}$(X) and state its cardinality.
5. Use a truth table to verify A$\Rightarrow$B $\Leftrightarrow$ $\neg$A$\vee$B. Which input row makes both sides false?
6. Let A=[0,2] and B=(1,3). Find A$\cap$B and A$\cup$B.
7. Negate: "For every integer n, $n^2 \geq 0$."
8. Let X={1,2,3,4} and A={1,4}. Find A^c relative to X.
9. Build a truth table for $\neg$(A$\vee$B). In which row is it true?
10. If |X|=3, how many elements does $\mathcal{P}$(X) have?
11. Translate "There is exactly one real x such that $x^2=4$" using quantifier notation.
12. Let A={1,2}, B={x,y,z}. List A$\times$B.
13. Are A$\Rightarrow$B and B$\Rightarrow$A equivalent for all A,B? Give one truth assignment that disproves equivalence.
14. Let X=$\mathbb{R}$ and A=[1,2]. Describe A^c as intervals.
15. Let P(x) mean "x is even", with domain $\mathbb{Z}$. Write "some integer is not even" symbolically.
16. Let A={1,2,3}, B={2,3,4}, C={3,4,5}. Find A$\cap$(B$\cup$C).
17. Classify each as a statement or not: (i) 4 is prime; (ii) x+1=2; (iii) 4 is lucky.
18. Use De Morgan's law to simplify $(A\cap B)^c$.
19. For A=0 and B=0, compare A$\Rightarrow$B with $\neg$A$\vee$B. Then use a truth table to verify $\neg(A\wedge B) \Leftrightarrow \neg A\vee\neg B$.
20. Let A={1,2,3,4} and B={2,4}. Is B$\subset$A? Is B$\subsetneq$A?
21. Negate: "There exists a real x such that $x^2<0$."
22. Let X={1,2,3,4,5}, A={1,2,3}, B={3,4}. Find $(A\cup B)^c$ relative to X.
23. Compare $\exists D \forall P$ R(D,P) with $\forall P \exists D$ R(D,P) in plain language.
24. For A={1,2} and B={2,3}, find A$\times$B and B$\times$A. State one ordered pair in only one of them.

Answers: see "Mixed practice" under Answers at the end.

Exam-style problems

Set a timer for about 20 minutes per problem. Write the reasoning, not just the final values.

1 · Logic for a sensor check

Three sensors report truth values A, B, and C. A majority vote is represented by $V=(A\wedge B)\vee(A\wedge C)\vee(B\wedge C)$.

1. Make a truth table for V and identify the rows where $V=1$.
2. Show that swapping any two sensor labels leaves V unchanged, using commutativity of $\wedge$ and $\vee$.
3. Negate the claim "For every sensor configuration with at most one report equal to 1, the majority output is 0." State the form of a counterexample.

Answers: see "Exam-style problems" under Answers at the end.

2 · Sets of admissible inputs

Let the universal set be $X=\{1,2,3,4,5,6,7,8\}$. Let A be the even elements of X, $B=\{1,2,3,4\}$, and $C=\{3,4,5,6\}$.

1. List A, $A\cap B$, and $A\cup B$.
2. Find $(B\cup C)^c$ and $B^c\cap C^c$ relative to X, and use the results to check a De Morgan identity.
3. Let $D=\{\text{red, blue}\}$. List $A\times D$ and give its cardinality.

Answers: see "Exam-style problems" under Answers at the end.

Slips in the notes

One statement in the lecture notes needs an edge-case qualification.

1. **Printed page 9:** the notes say that $A\times B$ is neither contained in A nor in B. If either factor is empty, then $A\times B=\varnothing$, and $\varnothing$ is a subset of both factors. The definition of the Cartesian product remains correct; the remark about containment is too broad for empty factors.

Self-check

Use day 0 to learn and practise. Tick a skill only in a later session, at least one day afterward, when you can show it without notes. Use the listed problem numbers as evidence.

- ☐ Evaluate all five logical connectives from input truth values. (1, 5, 9, 13, 19)
- ☐ Distinguish a statement from an open or undefined sentence. (3, 17)
- ☐ Prove equivalence with complete truth tables. (5, 19)
- ☐ Translate and negate quantified claims. (7, 11, 15, 21, 23)
- ☐ Compute unions, intersections, and differences from membership. (2, 6, 16)
- ☐ Use subset, proper subset, and equality definitions. (20)
- ☐ Find complements relative to an explicit universe. (8, 14, 22)
- ☐ Enumerate small power sets and count subsets. (4, 10)
- ☐ List Cartesian products and track coordinate order. (12, 24)
- ☐ Apply De Morgan's laws to statements and sets. (18, 19)
- ☐ Rebuild the listed derivations from definitions.
- ☐ Solve both multi-part problems and justify each step.

Answers

Start from memory

1. Only when A is true and B is false. §1.1
2. Yes. $A \vee B$ is true when at least one is true, including when both are true. §1.1
3. There exists an x for which P(x) is false: $\exists x\, \neg P(x)$. §1.1
4. Every element of A is also an element of B. §1.2
5. $A \times B$ contains ordered pairs (a,b) with $a \in A$ and $b \in B$; it is not the set {A,B}. §1.2

1.1 Propositional Logic

1. A=1, B=0. §1.1
2. Show that their output columns agree for every assignment. §1.1
3. $\exists x\, \neg P(x)$: at least one value fails the predicate. §1.1

1.2 Set Theory

1. Show both $A \subset B$ and $B \subset A$. §1.2
2. $A \cap B$ is a set of elements common to both; $A \times B$ is a set of ordered pairs with first and second components in the respective sets. §1.2
3. A universal set X containing A. §1.2

Explain it

1. The defined truth condition is "A is false or B is true." Only a true premise paired with a false conclusion violates it. §1.1
2. Equivalence means same output for every possible input; the table lists every input assignment. §1.1
3. A universal claim says every domain element works, so one failing element is enough to make it false. §1.1
4. $\exists D \forall P$ can use one dog for all people; $\forall P \exists D$ allows a different dog for each person. §1.1
5. "Outside A" means outside A but still within the chosen universe; changing the universe changes the complement. §1.2
6. $A \setminus B$ starts from A and removes elements of B; reversing the order changes both the candidates and what is removed. §1.2
7. The first coordinate must come from A and the second from B, so swapping the factors swaps coordinate roles. §1.2
8. Two sets are equal when they contain exactly the same elements, so membership of an arbitrary x captures equality. §1.2

Mixed practice

1. With A=1 and B=0, $\neg$A=0, A$\wedge$B=0, A$\vee$B=1, A$\Rightarrow$B=0, and A$\Leftrightarrow$B=0. The implication fails only on this true-to-false row; equivalence fails because the inputs differ. §1.1
2. Members in either set give A$\cup$B=$\{1,2,3,4\}$; the shared member gives A$\cap$B=$\{3\}$. Removing B from A leaves $\{1,2\}$, while removing A from B leaves $\{4\}$. §1.2
3. Yes. It is a declarative claim with a definite truth value, although that value is not known. §1.1
4. $\mathcal{P}$(X)=$\{\varnothing,\{a\},\{b\},\{a,b\}\}$ and $|\mathcal{P}(X)|=4=2^2$. §1.2
5. For rows (A,B)=(1,1),(1,0),(0,1),(0,0), both A$\Rightarrow$B and $\neg$A$\vee$B have outputs 1,0,1,1. They agree on every row, and both are false only for A=1,B=0. §1.1
6. A$\cap$B=(1,2]; A$\cup$B=[0,3). §1.2
7. There exists an integer n with $n^2<0$. This is false over $\mathbb{Z}$. §1.1
8. A^c=X\A=$\{1,2,3,4\}\setminus\{1,4\}=\{2,3\}$; the complement is taken within X. §1.2
9. It is true only when A=0 and B=0, because both must be false for $\neg$(A$\vee$B). §1.1
10. A three-element set has $2^3=8$ subsets: each element is either included or excluded independently. §1.2
11. $\exists!x\in\mathbb{R}$ such that $x^2=4$. The solutions are 2 and -2, so the claim as written is false. §1.1
12. $\{(1,x),(1,y),(1,z),(2,x),(2,y),(2,z)\}$. The first coordinate comes from A. §1.2
13. No. With A=1,B=0, A$\Rightarrow$B is false but B$\Rightarrow$A is true. §1.1
14. $(-\infty,1)\cup(2,\infty)$. The endpoints belong to A, so they are excluded from the complement. §1.2
15. $\exists x\in\mathbb{Z}$ such that $\neg$P(x), or "there is an odd integer." §1.1
16. B$\cup$C=$\{2,3,4,5\}$; intersecting with A gives $\{2,3\}$. §1.2
17. (i) Statement, false. (ii) Not a statement until x is assigned or quantified. (iii) Not a mathematical statement because "lucky" has no defined truth condition. §1.1
18. $(A\cap B)^c=A^c\cup B^c$: being outside both together means being outside at least one. §1.2
19. For A=B=0, both A$\Rightarrow$B and $\neg$A$\vee$B equal 1. For rows (1,1),(1,0),(0,1),(0,0), both $\neg$(A$\wedge$B) and $\neg$A$\vee\neg$B have values 0,1,1,1, so the columns match. §1.1
20. Yes, every element of B is in A. Yes, since $1\in A$ but $1\notin B$, so A has an element outside B. §1.2
21. For every real x, $x^2\geq 0$. §1.1
22. A$\cup$B=$\{1,2,3,4\}$; its complement in X is $\{5\}$. §1.2
23. The first says one D relates to every P. The second says each P has at least one related D, possibly different for each P. §1.1
24. A$\times$B=$\{(1,2),(1,3),(2,2),(2,3)\}$; B$\times$A=$\{(2,1),(2,2),(3,1),(3,2)\}$. For example, (1,2) is only in A$\times$B. §1.2

Exam-style problems

1. V=1 when at least two reports are 1: rows 111, 110, 101, 011. It is 0 for 100, 010, 001, 000. §1.1
2. For example, swapping A and B changes (A∧B) to (B∧A), which is equal by commutativity; it swaps the other two terms as well. The full disjunction is unchanged, and the same argument applies to any pair. §1.1
3. ∃ a configuration with at most one 1 but V=1. This cannot happen for the majority rule, so the original claim is true. §1.1

Exam-style problems

1. A={2,4,6,8}; A∩B={2,4}; A∪B={1,2,3,4,6,8}. §1.2
2. B∪C={1,2,3,4,5,6}, so $(B\cup C)^c$={7,8}. Also B^c={5,6,7,8} and C^c={1,2,7,8}; their intersection is {7,8}. Hence $(B\cup C)^c=B^c\cap C^c$. §1.2
3. A×D={(2,red),(2,blue),(4,red),(4,blue),(6,red),(6,blue),(8,red),(8,blue)}; it has 8 ordered pairs. §1.2

Personal study guide based on Mathematics for Engineers I lecture notes, Chapter 1 “Statements, Logic, and Sets” (printed pages 1–10), by Rhoslyn Coles and Cornelia Schneider. Written in my own words; no lecture pages or figures are reproduced. Not an official FAU publication; check course details against the current notes.